

# DEBIDUTTA BEHERA

## Software Engineer

Athagarh, Cuttack, Odisha, 754029, India

+91 7978213833 | [debidutta.db@gmail.com](mailto:debidutta.db@gmail.com) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## SUMMARY

Results-driven Software Engineer with hands-on experience designing, developing, and maintaining scalable software applications using Python, FastAPI, Django REST Framework, and React.js. Proven success collaborating with cross-functional teams to define and implement new features, while testing, debugging, and optimizing systems for performance, scalability, and reliability. Delivered 99.8% system uptime at DRDO's Integrated Test Range and consistently improved quality through automation and performance tuning. Eager to bring strong ownership and continuous improvement to a modern product engineering team.

## EDUCATION

|                                                                                                    |             |
|----------------------------------------------------------------------------------------------------|-------------|
| <b>Radhakrishna Institute of Technology and Engineering</b> B.Tech in<br>Computer Science (74.37%) | 2017 - 2021 |
| <b>Stewart Science College</b> +2 in<br>Science (52%)                                              | 2014 - 2017 |
| <b>S. S. V. M Tulasipur</b><br>Matriculation (57%)                                                 | 2013 - 2014 |

## WORK EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Apprentice Trainee — Software Developer &amp; Automation Engineer</b><br>Integrated Test Range (ITR), DRDO Gov. Of India   Chandipur, India                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 01/2025 - 01/2026 |
| <ul style="list-style-type: none"><li>Built and maintained Python-based backend services for mission-critical enterprise applications, achieving 99.8% system uptime through robust service design and ownership of the full feature lifecycle.</li><li>Automated UI testing protocols using Python Selenium and PyTest, reducing manual testing effort by 40% and deployment errors by 25% while improving release reliability.</li><li>Optimized frontend/backend Python scripts to eliminate runtime bottlenecks, delivering a 35% reduction in execution time through proactive performance profiling and debugging.</li><li>Participated in code reviews, authored technical documentation, and collaborated cross-functionally on internal software modules to identify improvements and implement new features.</li></ul> |                   |

## PROJECTS

### Cinema Seat Reservation System

- Python, Django REST Framework, React.js, TypeScript, PostgreSQL
- Architected RESTful backend endpoints with Django REST Framework to handle real-time booking transactions, seat-tier pricing logic, and data persistence in a PostgreSQL relational database.
- Built a dynamic, component-driven frontend using React.js and TypeScript with cross-device compatibility, real-time seat selection state management, and responsive UI rendering.
- Designed PostgreSQL schema with normalized tables for users, bookings, and pricing tiers, optimizing query performance for concurrent seat reservation transactions.

### Digital Attendance Tracker

- Python, Django, React.js, TypeScript, PostgreSQL, MySQL, HTML, CSS
- Developed a full-stack web portal using Django backend and React.js frontend with TypeScript, enabling role-based access control for administrators and employees.

- Implemented secure authentication, responsive form components, and a PostgreSQL/MySQL database schema that reduced manual attendance errors and streamlined HR operations.

#### **AWS Agreement RAG Chatbot**

- FastAPI, Streamlit, LangChain
- Developed an AI-powered chatbot for querying AWS agreement documents using natural language.
- Built a FastAPI backend for document processing, embeddings, and retrieval-based question answering.
- Created a Streamlit frontend for document upload and interactive chat.
- Integrated LangChain and vector search to deliver accurate, context-aware responses.

#### **AI-Powered Personal Assistant (JARVIS)**

- Python, SpeechRecognition, NLP
- Built a voice-activated assistant using Python NLP libraries to parse and execute system commands, implementing robust error-handling that improved command recognition accuracy by 20%.

#### **E-Commerce Frontend Web Application**

- React.js, TypeScript, HTML5, CSS3
- Built modern UI pages including product listing, cart, and checkout workflows.
- Implemented component-based architecture for scalable frontend development.
- Used API integration and async/await for product data fetching.
- Ensured cross-browser compatibility and mobile responsiveness.

#### **Automated Software Testing Framework**

- Python, Selenium, PyTest
- Engineered a cross-browser automated UI testing framework validating login workflows, user interactions, and form functionality across Chrome, Firefox, and Edge.
- Generated structured HTML test reports with automated screenshot capture on failure, reducing average debugging time significantly and enabling faster CI/CD feedback loops.

## **ACHIEVEMENTS**

---

- Strong understanding of frontend and backend fundamentals including component lifecycle, state management, and API handling.
- Built and deployed multiple frontend and full-stack projects using React and Python technologies.
- Continuously learning advanced frontend and backend engineering concepts and modern development practices.

## **SKILLS**

---

Python, FastAPI, Django REST Framework, RESTful APIs, React.js, JavaScript, TypeScript, HTML5, CSS3, Bootstrap, Tailwind CSS, PostgreSQL, MongoDB, Selenium, PyTest, Software Development, Debugging, Performance Optimization, Scalability, Reliability, Cross-functional Collaboration, Documentation, Object-Oriented Programming(OOP), JSON, Git, GitHub, Strimlit, Vite, Adobe Photoshop, Figma, Blender, Canva, Problem Solving, Team Managment, CI/CD Principles, Cross-Browser UI, Responsive Design, OR-Tools CP-SAT, UI/UX, schema design, joins, indexing, PHP, C++, SQL, JAVA, Node.JS, JQuery, HTTP protocols, status codes, request/response structures, No-SQL, Collaboration, Digital & AI Mindset, Technical Communication, logic, coding, system changes, AI, ML, Flask, .net, OpenAI APIs, LangChain, CrewAI, n8n, Prompt Engineering, AI Agent Development, SDLE, Basic Cloud Knowledge

## **LANGUAGES**

---

English (Fluent) · Hindi (Fluent) · Odia (Native)

## **HOBBIES & INTERESTS**

---

Coding and Building Personal Projects · Learning New Technologies · Participating in Hackathons · Solving Coding Challenges · Reading Tech Blogs and Articles · Exploring AI and Machine Learning Tools · Open-Source Contributions · Playing Chess · Cooking · Listening to Music